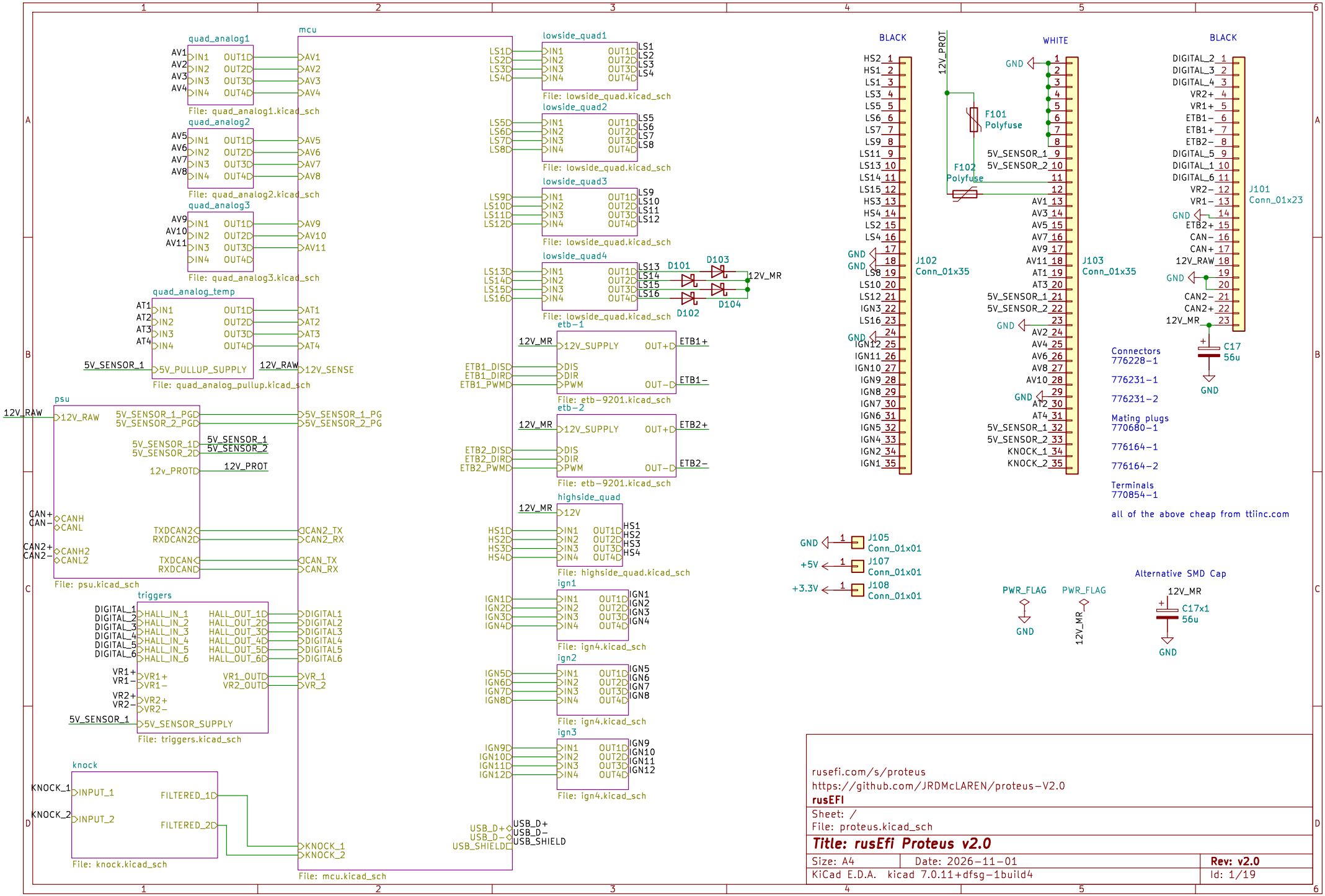
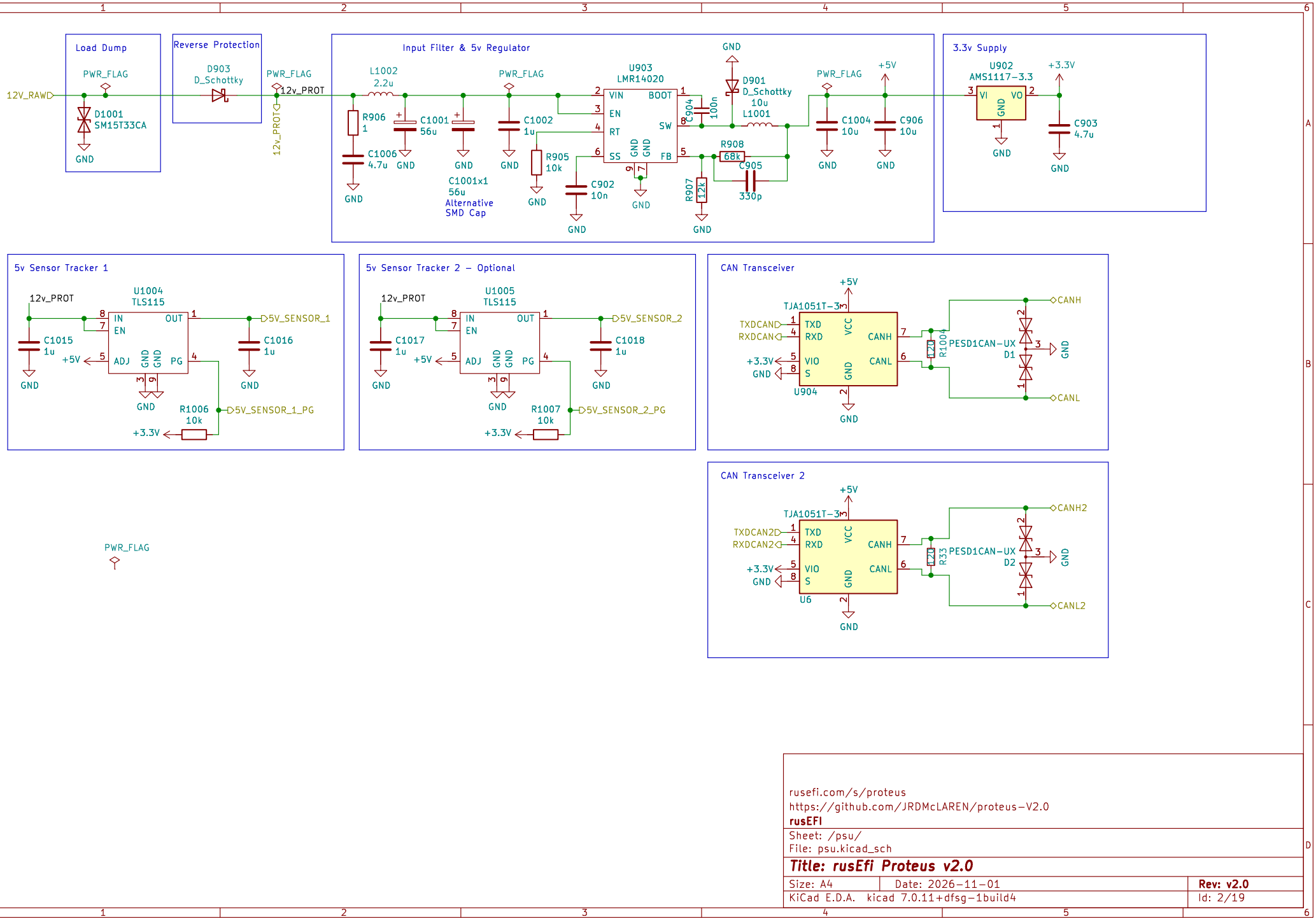
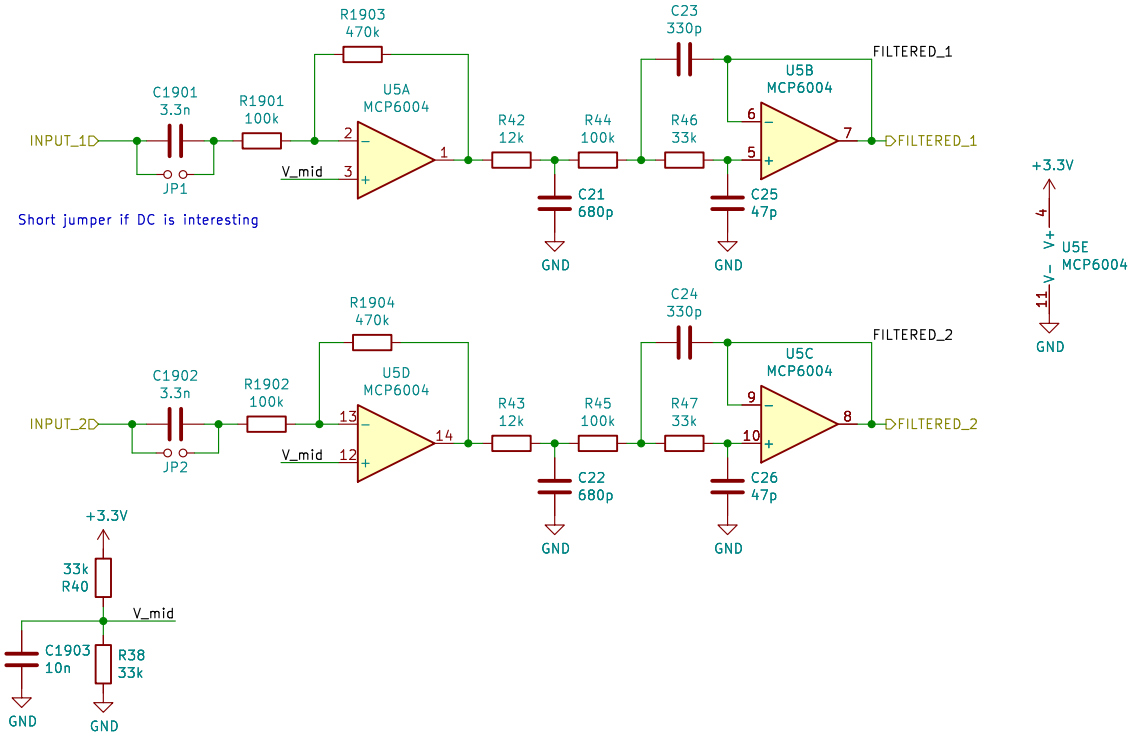
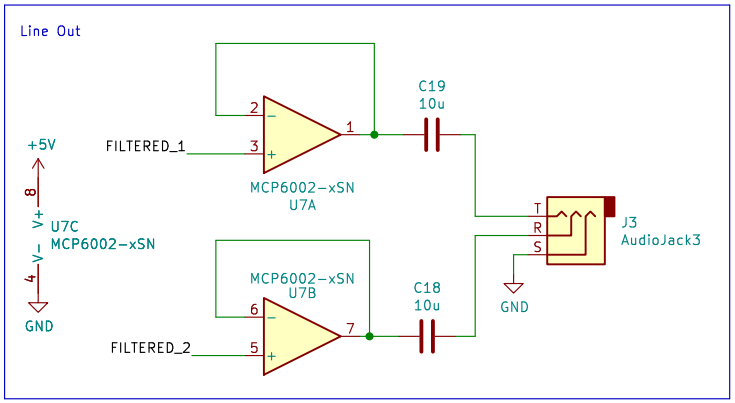

mcu
quad_analog1
File: quad_analog1.kicad_sch
quad_analog2
File: quad_analog2.kicad_sch
quad_analog3
File: quad_analog3.kicad_sch
quad_analog_temp
File: quad_analog_pullup.kicad_sch
psu
File: psu.kicad_sch
triggers
File: triggers.kicad_sch
knock
File: knock.kicad_sch
lowside_quad1
File: lowside_quad.kicad_sch
lowside_quad2
File: lowside_quad.kicad_sch
lowside_quad3
File: lowside_quad.kicad_sch
lowside_quad4
File: lowside_quad.kicad_sch
etb-1
File: etb-9201.kicad_sch
etb-2
File: etb-9201.kicad_sch
highside_quad
File: highside_quad.kicad_sch
ign1
File: ign4.kicad_sch
ign2
File: ign4.kicad_sch
ign3
File: ign4.kicad_sch
File: mcu.kicad_sch
BLACK
J102
Conn_01x35
WHITE
J103
Conn_01x35
BLACK
J101
Conn_01x23
F101
Polyfuse
F102
Polyfuse
12V_PROT
C17
56u
Connectors
776228-1
776231-1
776231-2
Mating plugs
770680-1
776164-1
776164-2
Terminals
770854-1
all of the above cheap from ttiinc.com
J105
Conn_01x01
J107
Conn_01x01
J108
Conn_01x01
PWR_FLAG
PWR_FLAG
Alternative SMD Cap
C17x1
56u
rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI
Sheet: /
File: proteus.kicad_sch
Title: rusEfi Proteus v2.0
Size: A4
Date: 2026-11-01
Rev: v2.0
KiCad E.D.A. kicad 7.0.11+dfsg-1build4
Id: 1/19


Load Dump
PWR_FLAG
12V_RAW
D1001
SM15T33CA
GND
Reverse Protection
D903
D_Schottky
PWR_FLAG
12v_PROT
12v_PROT
Input Filter & 5v Regulator
L1002
2.2u
R906
1
C1006
4.7u
C1001
56u
C1001x1
56u
Alternative
SMD Cap
PWR_FLAG
C1002
1u
R905
10k
C902
10n
U903
LMR14020
VIN
EN
RT
SS
BOOT
SW
FB
GND
C904
100n
D901
D_Schottky
10u
L1001
R908
68k
C905
330p
R907
12k
PWR_FLAG
C1004
10u
C906
10u
+5V
3.3v Supply
U902
AMS1117-3.3
VI
VO
GND
+3.3V
C903
4.7u
5v Sensor Tracker 1
12v_PROT
U1004
TLS115
IN
EN
OUT
ADJ
GND
PG
C1015
1u
C1016
1u
5V_SENSOR_1
R1006
10k
5V_SENSOR_1_PG
+5V
+3.3V
5v Sensor Tracker 2 – Optional
12v_PROT
U1005
TLS115
C1017
1u
C1018
1u
5V_SENSOR_2
R1007
10k
5V_SENSOR_2_PG
CAN Transceiver
TJA1051T-3
TXD
RXD
VCC
CANH
CANL
VIO
S
TXDCAN
RXDCAN
R1004
120
PESD1CAN-UX
D1
CANH
CANL
U904
CAN Transceiver 2
TXDCAN2D
RXDCAN2
R33
CANH2
CANL2
U6
PWR_FLAG
rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI
Sheet: /psu/
File: psu.kicad_sch
Title: rusEfi Proteus v2.0
Size: A4
Date: 2026-11-01
Rev: v2.0
KiCad E.D.A. kicad 7.0.11+dfsg-1build4
Id: 2/19



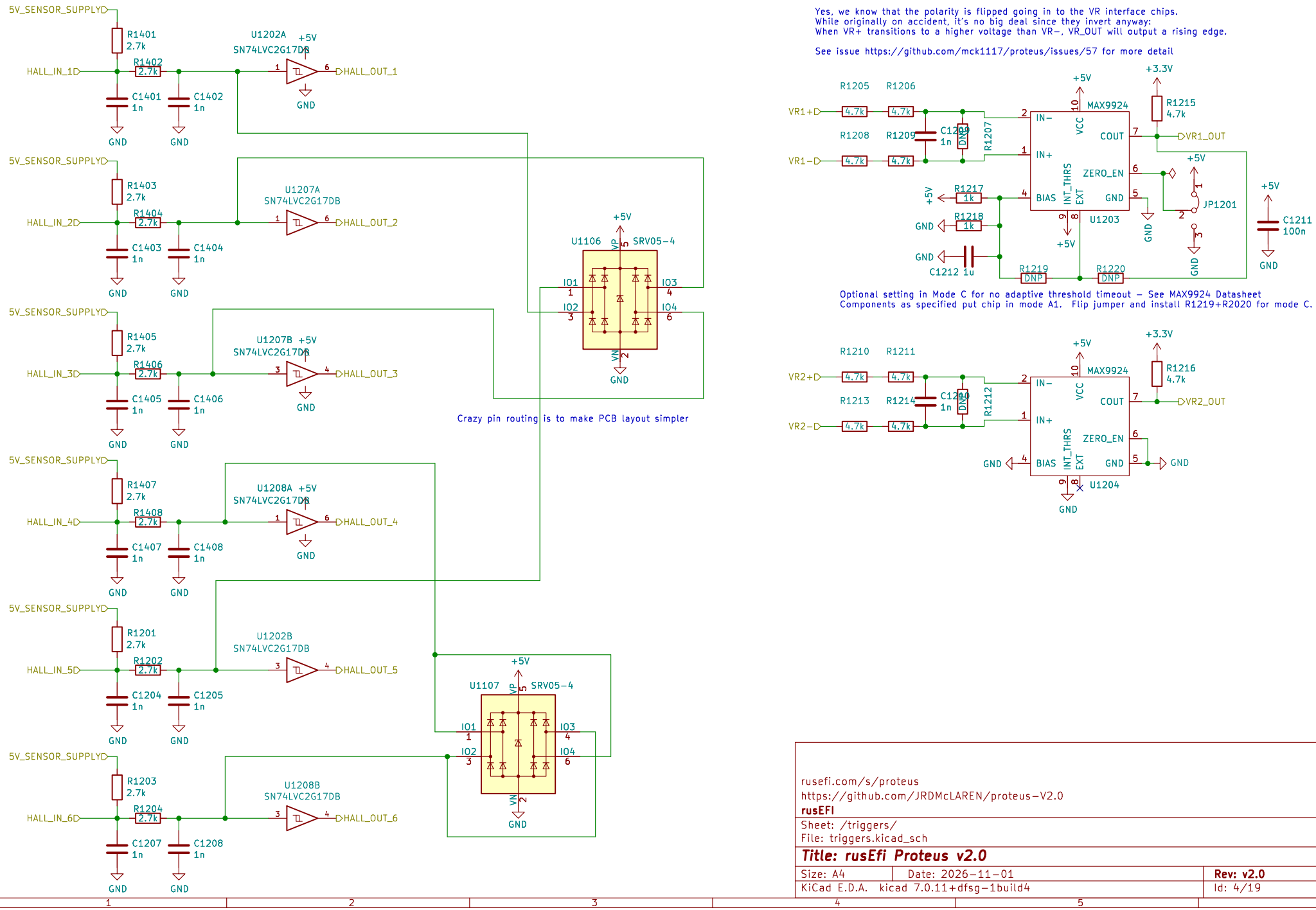


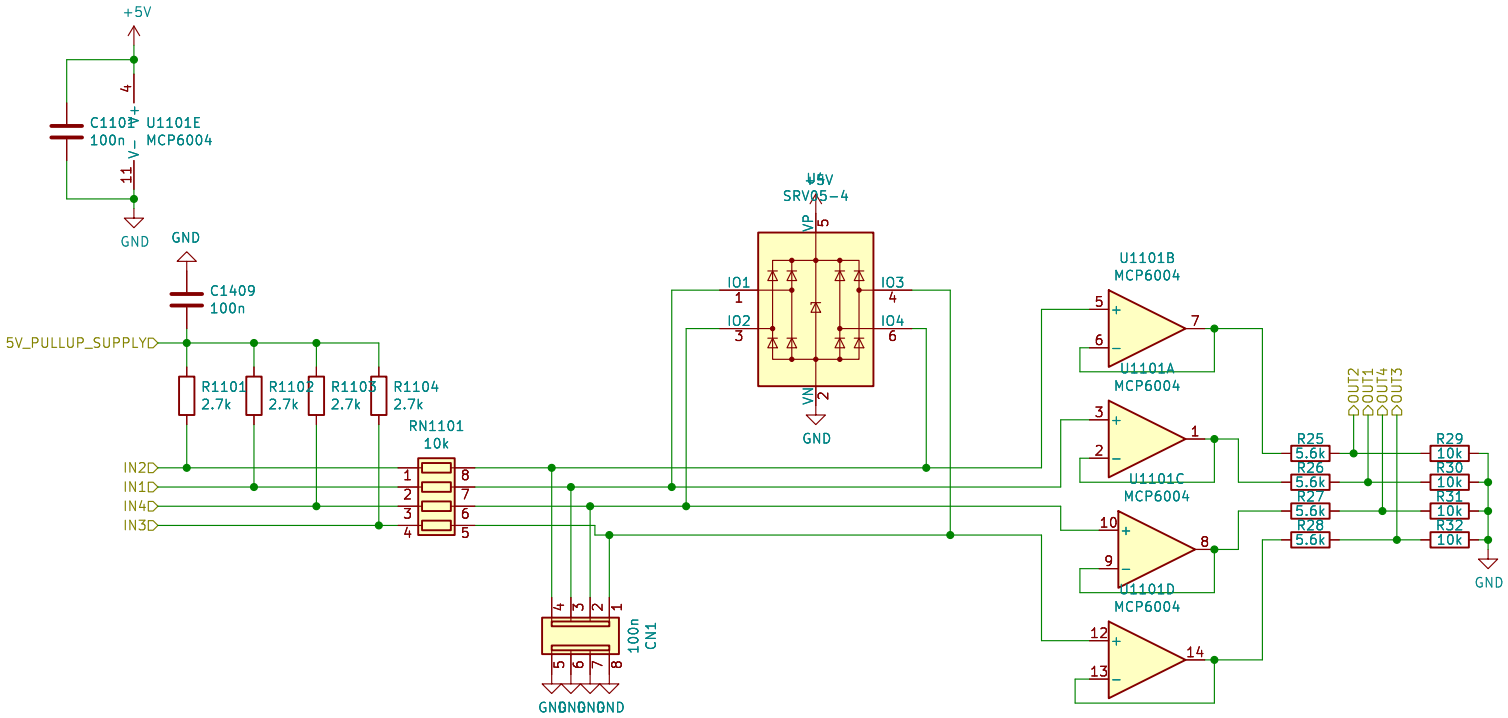
rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /knock/
File: knock.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 3/19


5V_SENSOR_SUPPLY
R1401 2.7k
R1402 2.7k
HALL_IN_1
C1401 1n
C1402 1n
GND
U1202A SN74LVC2G17DB
+5V
HALL_OUT_1
R1403 2.7k
R1404 2.7k
HALL_IN_2
C1403 1n
C1404 1n
U1207A SN74LVC2G17DB
HALL_OUT_2
R1405 2.7k
R1406 2.7k
HALL_IN_3
C1405 1n
C1406 1n
U1207B SN74LVC2G17DB
HALL_OUT_3
R1407 2.7k
R1408 2.7k
HALL_IN_4
C1407 1n
C1408 1n
U1208A SN74LVC2G17DB
HALL_OUT_4
R1201 2.7k
R1202 2.7k
HALL_IN_5
C1204 1n
C1205 1n
U1202B SN74LVC2G17DB
HALL_OUT_5
R1203 2.7k
R1204 2.7k
HALL_IN_6
C1207 1n
C1208 1n
U1208B SN74LVC2G17DB
HALL_OUT_6
U1106 SRV05-4
U1107 SRV05-4
VP
VN
IO1
IO2
IO3
IO4
Crazy pin routing is to make PCB layout simpler
Yes, we know that the polarity is flipped going in to the VR interface chips.
While originally on accident, it's no big deal since they invert anyway:
When VR+ transitions to a higher voltage than VR−, VR_OUT will output a rising edge.
See issue https://github.com/mck1117/proteus/issues/57 for more detail
VR1+
VR1−
R1205 4.7k
R1206 4.7k
R1208 4.7k
R1209 4.7k
C1209 1n DNP
R1207
MAX9924
U1203
IN−
IN+
VCC
COUT
ZERO_EN
BIAS
INT_THRS
EXT
R1215 4.7k
VR1_OUT
R1217 1k
R1218 1k
C1212 1u
JP1201
C1211 100n
R1219 DNP
R1220 DNP
Optional setting in Mode C for no adaptive threshold timeout – See MAX9924 Datasheet
Components as specified put chip in mode A1. Flip jumper and install R1219+R2020 for mode C.
VR2+
VR2−
R1210 4.7k
R1211 4.7k
R1213 4.7k
R1214 4.7k
C1210 1n DNP
R1212
U1204
R1216 4.7k
VR2_OUT
+3.3V
rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI
Sheet: /triggers/
File: triggers.kicad_sch
Title: rusEfi Proteus v2.0
Size: A4
Date: 2026-11-01
Rev: v2.0
KiCad E.D.A. kicad 7.0.11+dfsg-1build4
Id: 4/19

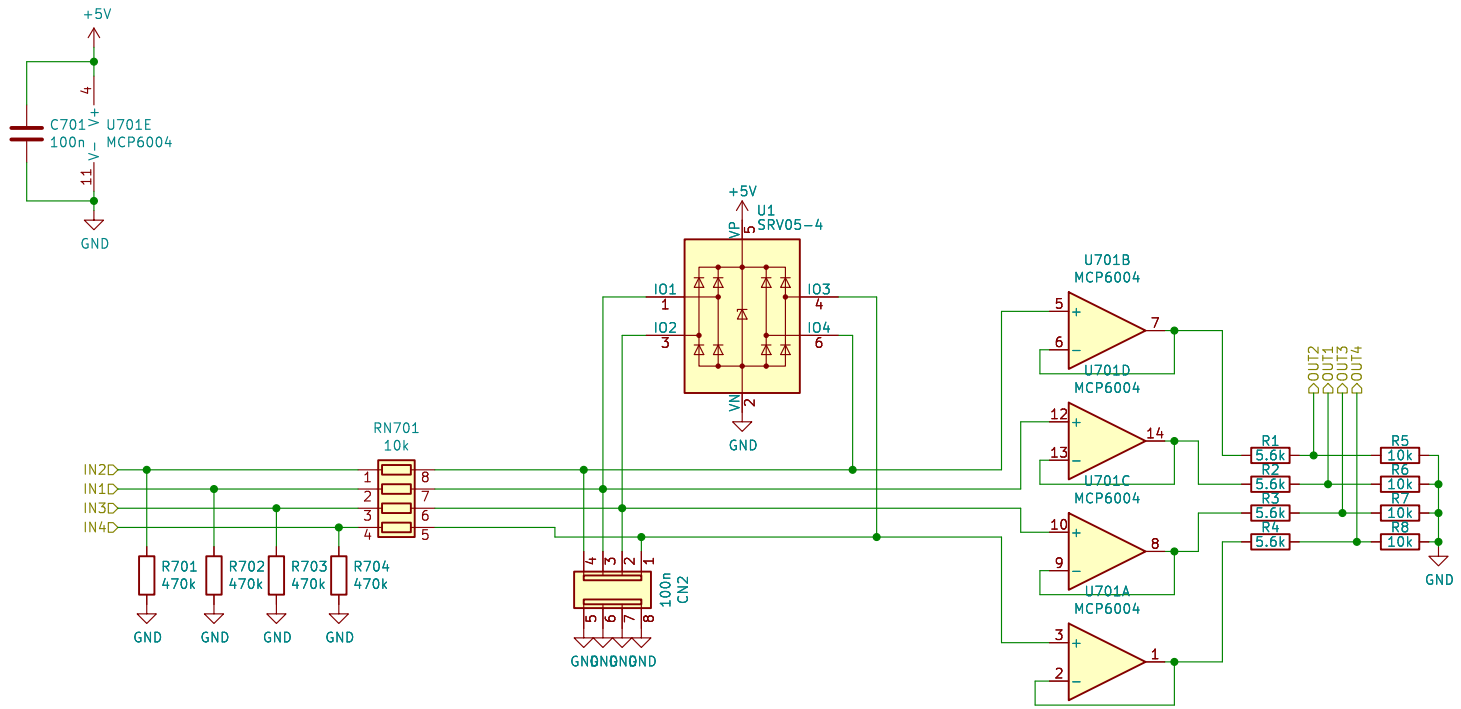


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /quad_analog_temp/
File: quad_analog_pullup.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 5/19

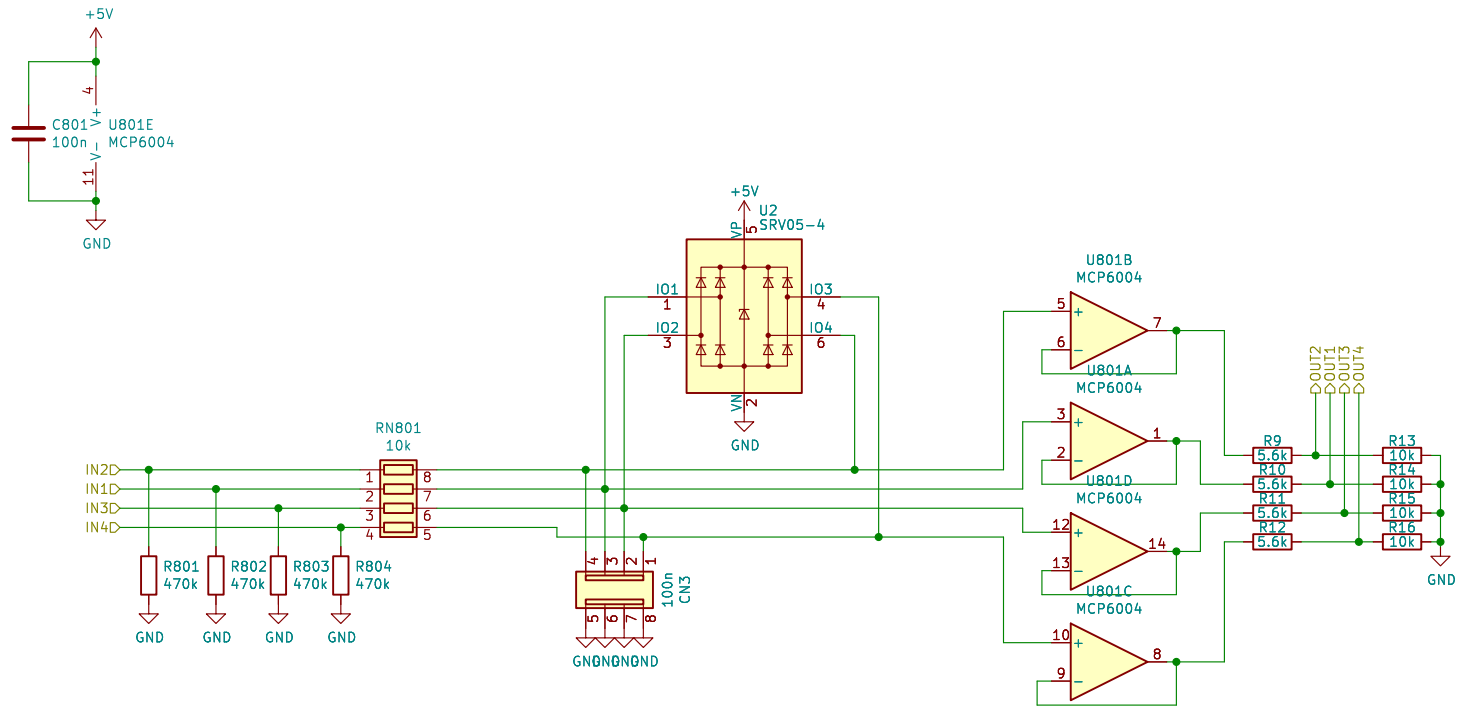


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /quad_analog1/
File: quad_analog1.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 6/19

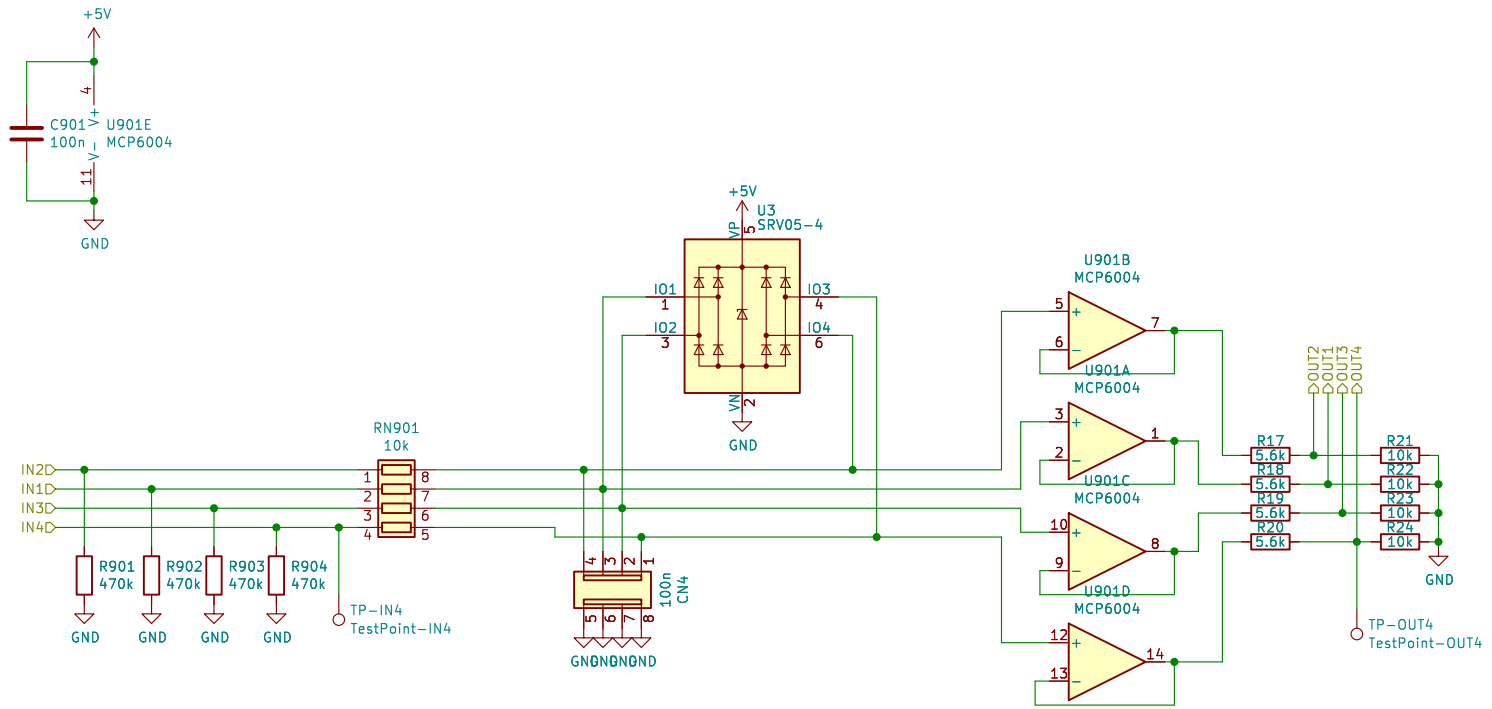


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /quad_analog2/
File: quad_analog2.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 7/19

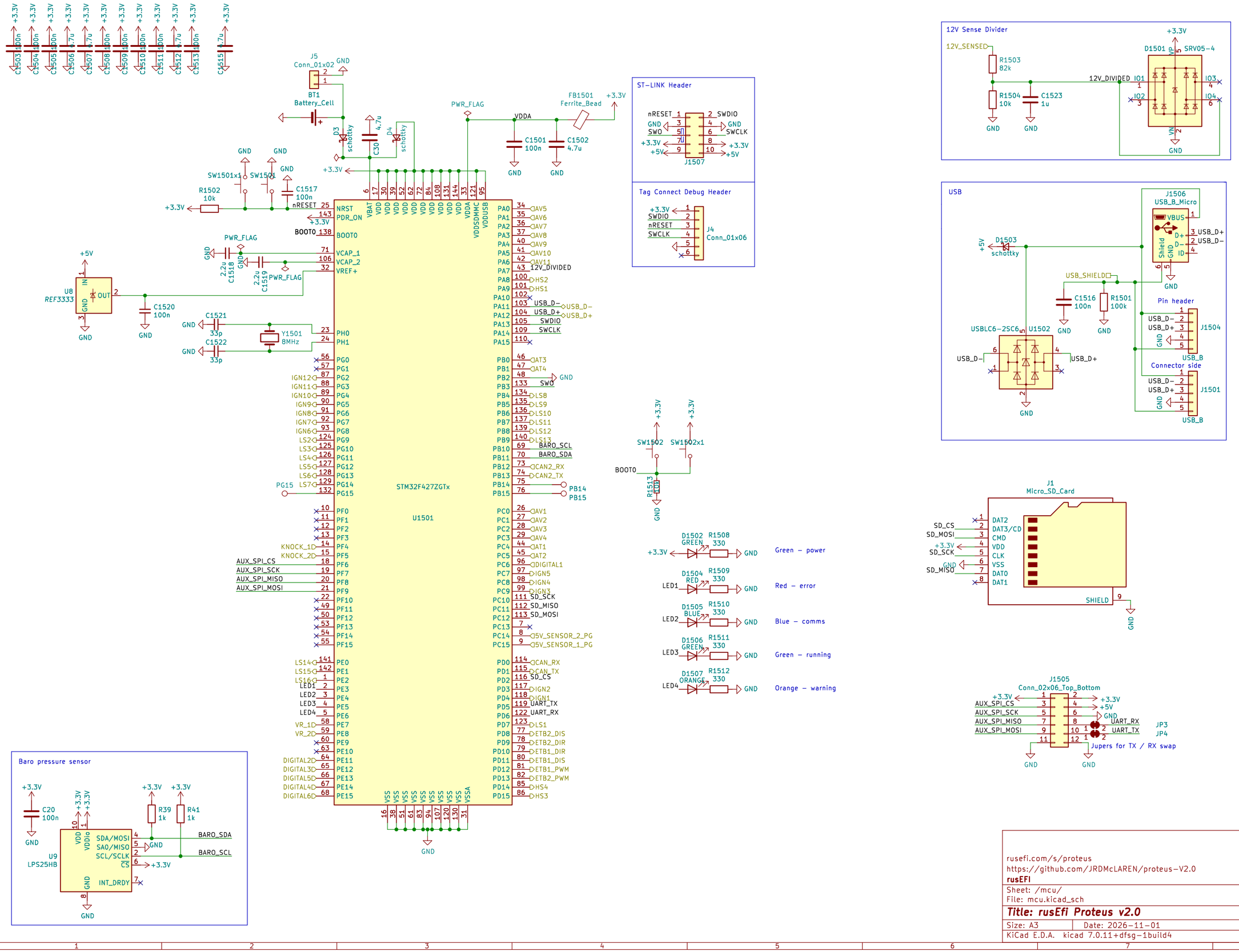


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /quad_analog3/
File: quad_analog3.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 8/19

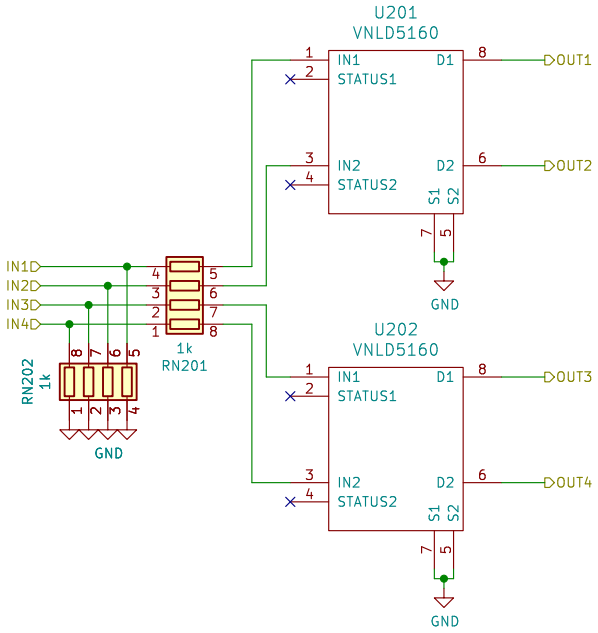


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /mcu/
File: mcu.kicad_sch

Title: rusEfi Proteus v2.0

Size: A3	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 9/19



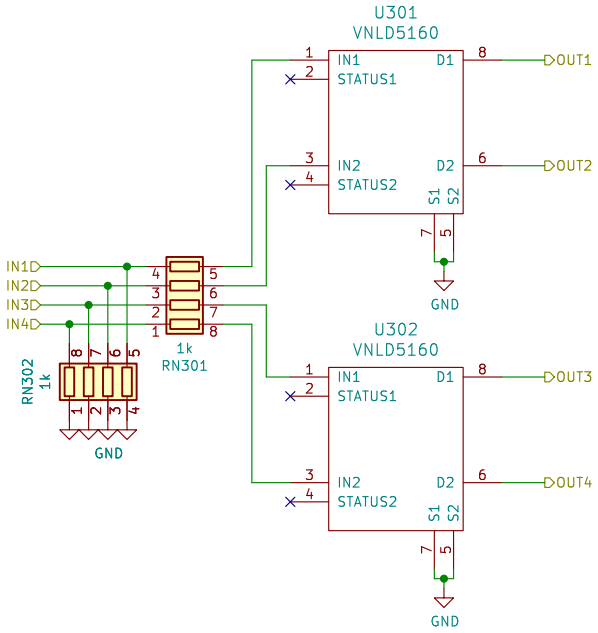
Locate output caps near connector pin

rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /lowside_quad1/
File: lowside_quad.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 10/19



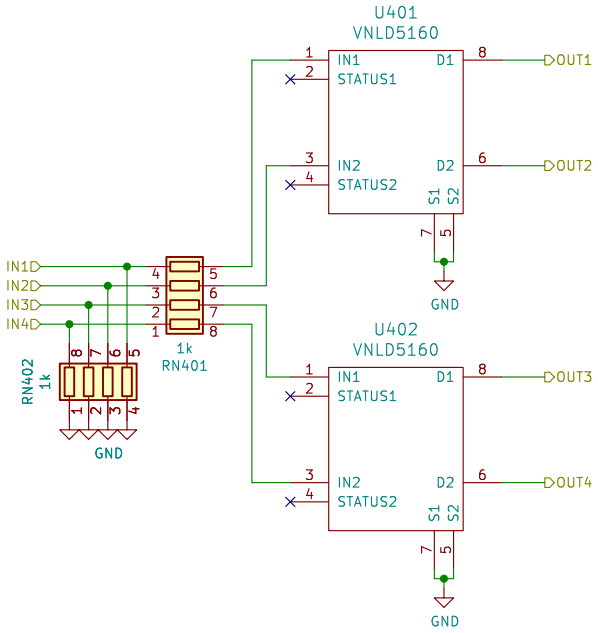
Locate output caps near connector pin

rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /lowside_quad2/
File: lowside_quad.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 11/19



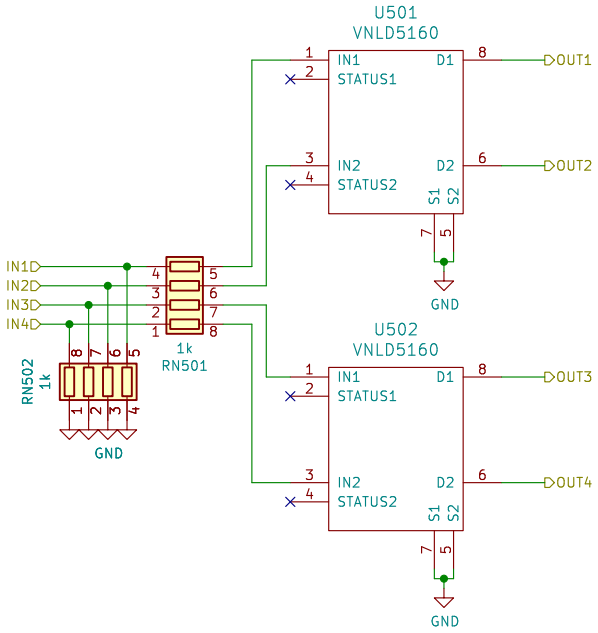
Locate output caps near connector pin

rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /lowside_quad3/
File: lowside_quad.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 12/19



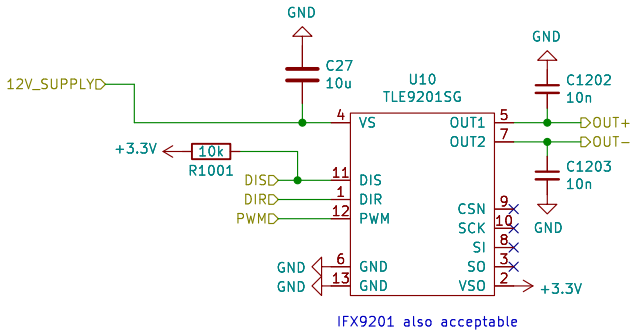
Locate output caps near connector pin

rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /lowside_quad4/
File: lowside_quad.kicad_sch

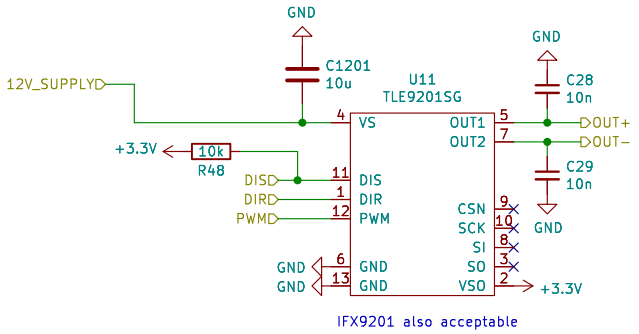
Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 13/19



rusefi.com/s/proteus	
https://github.com/JRDMcLAREN/proteus-V2.0	
rusEFI	
Sheet: /etb-1/	
File: etb-9201.kicad_sch	
Title: rusEfi Proteus v2.0	
Size: A4	Date: 2026-11-01
KiCad E.D.A. kicad 7.0.11+dfsg-1build4	Id: 14/19

Rev: v2.0

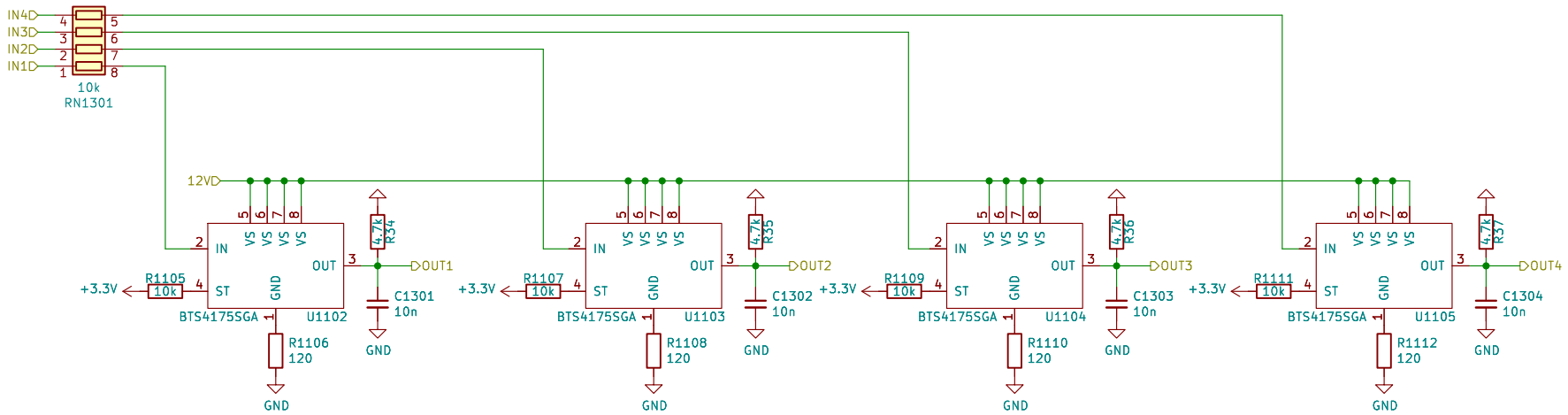


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /etb-2/
File: etb-9201.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 15/19

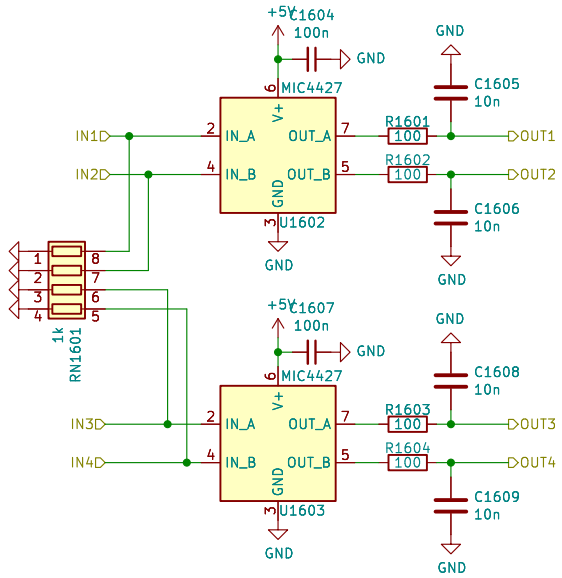


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /highside_quad/
File: highside_quad.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 16/19

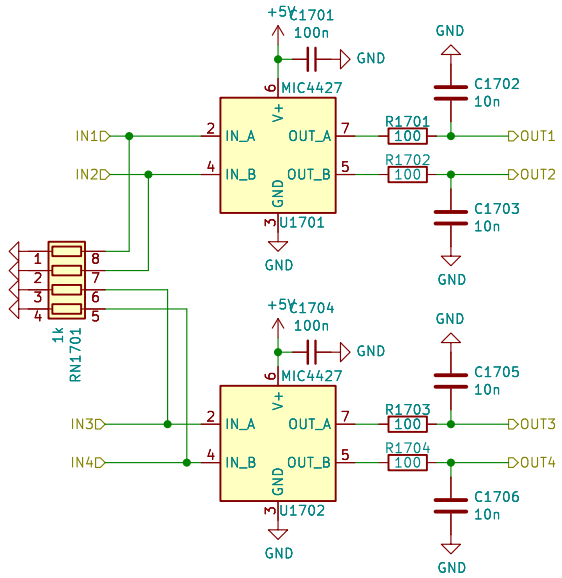


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /ign1/
File: ign4.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 17/19

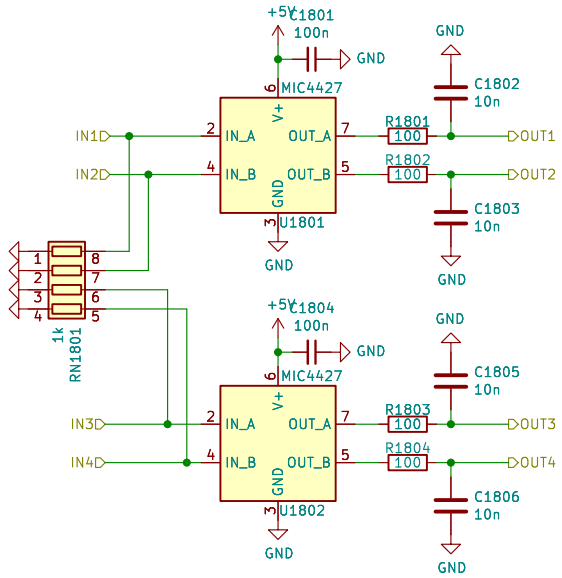


rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /ign2/
File: ign4.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 18/19



rusefi.com/s/proteus
https://github.com/JRDMcLAREN/proteus-V2.0
rusEFI

Sheet: /ign3/
File: ign4.kicad_sch

Title: rusEfi Proteus v2.0

Size: A4	Date: 2026-11-01	**Rev: v2.0**
KiCad E.D.A. kicad 7.0.11+dfsg-1build4		Id: 19/19